Jatismar Saha

PhD Researcher | Lead-free Perovskite Photovoltaics (PV) | Thin Films | Semiconductor Device Simulation (TCAD)

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Google Scholar: scholar.google.com/citations?user=4rPTNDYAAAAJ

PROFILE

PhD researcher at IIT Guwahati working on lead-free, tin-based perovskite solar cells; currently a JST-LOTUS Fellow at the National Institute for Materials Science (NIMS), Japan. Hands-on perovskite solar cell fabrication in an inert-atmosphere glovebox, thermal evaporation and FESEM at NIMS, with XRD training in progress; cleanroom process training at IIT Guwahati in PECVD, RF sputtering and oxidation / diffusion furnace. Three peer-reviewed 2025 journal papers on tunnel FET simulation in Synopsys Sentaurus TCAD, including heavy-ion radiation effects. Open to research collaborations.

SKILLS

Photovoltaics and thin films	Perovskite solar cell fabrication, lead-free (tin-based) perovskites, photovoltaics (PV), inert-atmosphere glovebox processing, spin coating, thermal evaporation, RF sputtering, plasma-enhanced chemical vapor deposition (PECVD), oxidation / diffusion furnace, cleanroom processing
Materials characterization	Field-emission scanning electron microscopy (FESEM), X-ray diffraction (XRD, training in progress), UV-Visible spectroscopy (UV-Vis)
Device simulation	Synopsys Sentaurus TCAD, semiconductor device modeling, tunnel field-effect transistors (TFET), silicon-on-insulator (SOI), radiation effects (heavy-ion), interface trap charges
Design and programming	Python, Cadence Virtuoso, Xilinx Vivado, KiCad, Eagle PCB, Arduino

RESEARCH EXPERIENCE

JST-LOTUS Fellow May 2026 – Present

National Institute for Materials Science (NIMS), Tsukuba, Japan

- Fabricate perovskite solar cells in an inert-atmosphere glovebox and deposit device layers by thermal evaporation.
- Characterize film morphology by FESEM; in XRD training for crystal structure and phase analysis.

Doctoral Researcher Jul 2025 – Present

School of Energy Science and Engineering, Indian Institute of Technology (IIT) Guwahati, India

- PhD research on lead-free, tin-based perovskite solar cells: fabrication, characterization and device modeling.
- Cleanroom process training: oxidation / diffusion furnace, PECVD, RF sputtering.

MTech Research, Semiconductor Device Simulation Aug 2023 – May 2025

National Institute of Technology (NIT) Silchar, India

- Simulated n-type step tunneling path TFETs and silicon-on-insulator (SOI) n-p-n double-gate TFETs in Synopsys Sentaurus TCAD.
- Simulated the effects of heavy-ion strikes, interface trap charges and temperature on TFET performance.
- Published 3 peer-reviewed journal papers in 2025 (Elsevier, Springer, Wiley).

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **J. Saha**, B. Choudhuri, R. Saha, "Radiation effects in silicon-on-insulator n-p-n double-gate TFET: a simulation study," *physica status solidi (a)*, vol. 222, no. 24, e202500615, 2025. doi: 10.1002/pssa.202500615
2. **J. Saha**, M. P. Gogoi, S. Tiwari, B. Choudhuri, R. Saha, "Investigating the effects of heavy-ion radiation on n-type step tunnelling path TFET," *Micro and Nanostructures*, vol. 207, 208269, 2025. doi: 10.1016/j.micrna.2025.208269
3. **J. Saha**, M. P. Gogoi, B. Choudhuri, R. Saha, "Investigating the effects of interface trap charges and temperature on n-type step tunneling path TFET," *Analog Integrated Circuits and Signal Processing*, vol. 125, no. 1, 17, 2025. doi: 10.1007/s10470-025-02505-x

BOOK CHAPTER

4. **J. Saha**, R. Choudhuri, D. Kalita, J. Swarnakar, "Development of a smart railway platform system," in *Futuristic Trends in Network & Communication Technologies*, IIP Series, vol. 3, pp. 186–196, 2023. doi: 10.58532/V3BINC3P6CH5

HONORS AND FELLOWSHIPS

JST-LOTUS Fellowship, Japan Science and Technology Agency (JST) 2026

EDUCATION

PhD, Energy Science and Engineering Jul 2025 – Present
Indian Institute of Technology (IIT) Guwahati, India · Supervisor: Dr. Ajay Singh

MTech, Microelectronics and VLSI Design Aug 2023 – May 2025
National Institute of Technology (NIT) Silchar, India · CGPA 7.48/10

BTech, Electronics and Communication Engineering Aug 2019 – Jul 2023
North-Eastern Hill University (NEHU), Shillong, India · CGPA 7.25/10

INTERNSHIPS

PCB design for an indoor drone, Indian Institute of Technology (IIT) Guwahati · schematic and layout in KiCad and Eagle PCB Oct 2024 – Jun 2025

Fixed-wing single-rotor aircraft development, IIT Guwahati 2023

Hexacopter drone development, IIT Guwahati · Mission Planner Aug – Sep 2022

PCB fabrication and 3D printing, IIT Guwahati May – Jun 2022

LANGUAGES

English, Hindi, Assamese, Bengali; Japanese (beginner, currently taking classes)

REFERENCES

Available on request.